

Tide retrospective: unrestricted exponential bounds the scalar layer of the numerator stage

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Abstract

Mathlib’s `Real.exp_bound` requires $|x| \leq 1$, and on the mesoscopic window the exponent correction reaches $\sim 1/\sqrt{q}$: it is small relative to the quadratic, never absolutely. The architecture consult confirmed that the right fix is a one-time unrestricted remainder — for every real x ,

$$\left| e^x - \sum_{i=0}^N \frac{x^i}{i!} \right| \leq \frac{|x|^{N+1} e^{|x|}}{(N+1)!},$$

with the q -power later recovered from the multinomial fact $|A(q, z)| \leq q \sum_s |V_s(z)|$ on $q \leq 1$ rather than from absolute smallness — and that a second, smaller localization window would cost more (its own tail theorems, annulus estimates) while still failing to bound the Peano remainder absolutely at $N = 0$. This tide delivers that remainder, the perturbation bound $|e^{-(A+\delta)} - e^{-A}| \leq |\delta| e^{|A|+|\delta|}$ shaped for Gaussian absorption, and the graded polynomial tail bound with a z -uniform indexing bound through degree N^2 .

1 Setting

Stage 5b of the forward-expansion programme: the scalar/algebraic layer, kept free of domain and measure structure per the consult, so that the single q -uniform Gaussian majorant of the next tide is the only place where domain-specific uniformity concentrates. Numerical checks preceded the Lean: the remainder inequality at twelve (x, N) pairs including $x = 5$, and the tail bound at the stage-4 test coefficients.

2 The thing formalised

The remainder proof is the series argument made literal: `real_exp_eq_tsum` identifies e^x with its power series (through `NormedSpace.expSeries_div_hasSum_exp`); the partial sum splits off by `Summable.sum_add_tsum_nat_add`; the shifted tail is compared termwise using $(N+1)!i! \leq (N+1+i)!$ (from Mathlib’s divisibility fact) and collapses to $\frac{|x|^{N+1}}{(N+1)!} e^{|x|}$ by `tsum_mul_left`. The perturbation bound reduces to $|e^y - 1| \leq |y|e^{|y|}$, proven by two `nlinarith` calls from `Real.add_one_le_exp` at $\pm y$. The tail bound mirrors stage 4’s polynomial tail step, but with the indexing bound $N^2 + 1$ from `natDegree_gradedExpPoly_le` instead of the polynomial’s own degree — the consult’s point being that after pointwise

instantiation the degree would depend syntactically on the point, poisoning every later uniformity statement.

3 Where progress stalled

Two deterministic whnf heartbeat timeouts (a catalogued class): the `NormedSpace.exp` instance unification on $\mathbb{R}$, and the `tsum` comparison chain. Both cleared by the documented `maxHeartbeats` bump; the first also needed the `HasSum`-based route, because rewriting with `exp_eq_tsum_div` dies in the same way at any budget tried. Two renames-and-shapes: `tsum_le_tsum` is now dot-notation on the left summability, and `simpα [Real.norm_eq_abs]` over-normalizes (it pushes the absolute value inside division and power, breaking the $\sum' |f_i|$ shape) — on $\mathbb{R}$ the norm lemmas apply directly by definitional equality.

4 Mathlib lookup versus new mathematics

Lookup: `NormedSpace.expSeries_div_hasSum_exp`, `Real.summable_pow_div_factorial`, `Summable.sum_add_tsum_nat_add`, `summable_nat_add_iff`, `norm_tsum_le_tsum_norm`, `tsum_mul_left`, `Nat.factorial_mul_factorial_dvd_factorial_add`, `Polynomial.natDegree_sum_le_of_forall_le`, `Polynomial.natDegree_pow_le`, `Polynomial.natDegree_C_mul_le`. New: nothing — these are textbook estimates; the value is having them stated in the exact shapes the majorant proof will consume.

5 What the next tide inherits

Stage 5c-pre, the domain absorption package (consult items 1–3): the T_2/H bridge with a named positive Gaussian rate, arbitrary-small Peano control on the window (`eventually_abs_scaledRem_le` from `taylorPeano` plus `smul_mem_ball_of_mesoscopic`), and the absorption of the absolute corrections into $e^{-\gamma\|z\|^2}$. Then the majorant, DCT, tail removals, and `numerator_hasExpansion`.